

BALA KARTHIK REDDY MARAMREDDY

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SUMMARY

- Full Stack Developer and Software Engineer with **4+ years of experience** designing, developing, and supporting scalable enterprise applications across **e-commerce, healthcare, research, and SaaS platforms**, leveraging **Java, Spring Boot, C#, ASP.NET Core, React, TypeScript, Python, and Node.js**.
- Proven expertise building **Microservices, Distributed Systems, REST/SOAP/GraphQL APIs, Event-Driven Architectures, ERP Integrations, and AI-powered applications**, including **LLM-based AI Agents, RAG pipelines, Prompt Engineering, and workflow automation**, delivering high-performance solutions for thousands of users.
- Hands-on experience with **AWS, Azure, Docker, MongoDB, MySQL, PostgreSQL, CI/CD Pipelines, Unit Testing, System Design, and Cloud-Native Development**, collaborating in Agile environments to deliver secure, scalable, and production-ready software solutions.

SKILLS

Programming Languages: Java, C#, Python, PHP, JavaScript, TypeScript, SQL, Bash, HTML5, CSS3, JSON, XML

Frontend: React, Angular, Redux, Next.js, Vue.js, jQuery, Bootstrap, Tailwind CSS, D3.js, Ajax, i18n (Internationalization)

Backend & APIs: Spring Boot, Spring MVC, ASP.NET Core, Entity Framework Core, Laravel, Node.js, Express.js, FastAPI, REST, SOAP, GraphQL, Hibernate, Flask, Django, OAuth 2.0, JWT

Databases: PostgreSQL, MySQL, SQL Server, MongoDB, SQLite, Embedded Databases, Redis

Architecture & Software Engineering: Microservices, Distributed Systems, Event-Driven Architecture, System Design, Design Patterns, SOLID Principles, OOP, Multi-Tenant Architecture, Scalability, ERP Integrations

Cloud Services: AWS (EC2, S3, RDS, Lambda, IAM, CloudFormation), Azure App Services, Azure DevOps, Docker, Kubernetes, Jenkins

Testing & QA: JUnit, NUnit, Jest, Unit Testing, Integration Testing, Debugging, Performance Optimization

Tools & Platforms: Git, Bitbucket, GitHub, VS Code, IntelliJ IDEA, DBeaver, ServiceNow, Maven, Postman, DBGate, Laragon, FreshService, ClickUp, Jira, Figma, Firebase, Kafka, Putty, WinSCP, Microsoft Teams, Slack, NetSuite, n8n

Healthcare & Standards: HL7, FHIR, HIPAA Compliance, ASP.NET Identity, Role-Based Access Control

DevOps & Practices: DevOps, CI/CD Pipelines, Unit Testing, Code Review, Pair Programming, Version Control, Systems Integration, Automation, Agile, Scrum, API Gateway

AI Tools: LLM, AI Agents, RAG (Retrieval-Augmented Generation), Vector Embeddings, ChatGPT, GitHub Copilot, Claude Sonnet, Prompt Engineering, Generative AI, MCP

EXPERIENCE

TOGA TECHNOLOGY (AGILANT SOLUTIONS INC.), ILLINOIS | FULL STACK DEVELOPER

MARCH 2025 - PRESENT

TOGA Platforms (Retail, Desk, View, IQ, Commerce, Supply, Hub)

- Designed and implemented **persona-based access control** for **TOGa Commerce** using **React, TypeScript, PHP, and Node.js**, building role-driven UI flows that dynamically control feature visibility, workflows, and data access across **8+ roles** and **20+ personas**, backed by **MySQL** schema design and **multi-tenant architecture** serving **85,000+ daily users**.
- Implemented backend integrations and a Retrieval-Augmented Generation (RAG) pipeline for Talos, an internal **LLM-powered AI agent** built using **Python, FastAPI, and Claude LLMs**, integrated with TOGa Desk and TOGa View, enabling automated ticket creation, real time updates, and programmatic resolution, reducing manual contact center effort by **30%** and improving response times.
- Built a document ingestion and embedding pipeline plus **prompt engineering** for Talos' conversational chat interface over internal company policy and HR documentation, achieving **80% response accuracy** and letting employees resolve company-rule questions directly through Talos instead of contacting HR.
- Contributed to scaling **React frontend and backend services** for **TOGa Commerce**, optimizing order placement kits, persona-based flows, and tracking workflows for a platform used by **85,000+ daily users**, **improving overall application responsiveness by 20%**.
- Developed and maintained end-to-end systems integrations between Freshservice and TOGa Desk using **REST, SOAP, and GraphQL APIs**, enabling fault-tolerant communication and automated workflow execution through webhook-driven event processing.
- Led feature development and enhancements across **TOGa** platforms following **Agile** and **Scrum** methodologies, providing **team leadership** and **mentoring** junior developers and ensuring smooth post-release production stabilization.
- Implemented client-specific UI logic for **10+ client onboardings** in **TOGa View** using a multi-tenant architecture, enabling configurable layouts, workflows, and data visibility.
- Designed scalable **MySQL** database schemas and asynchronous background processing workflows supporting **ERP and NetSuite integrations**, improving data consistency across **Distributed Systems** and supporting high-volume transactional workloads.
- Built and enhanced **React + TypeScript** UI components, applying lazy loading, component reusability, and conditional rendering to improve **web performance optimization** and support **cross-browser development**.
- Built a **multi-language switcher** for **TOGa Commerce** using **React** and **i18n**, enabling dynamic switching across **5 languages** for a diverse user base, improving accessibility, **cross-browser** and **cross-device** compatibility.
- Designed and implemented 20+ automated client reports using background workers, cron schedules, and **Event-Driven Architecture** patterns, generating scheduled email reports while maintaining system reliability under concurrent workloads.

- Actively supported production systems through **Bash** scripting and log analysis on **Linux** EC2 instances, backend fixes, database debugging, and API validation using **Postman**, conducted **code reviews** and **pair programming** sessions, managed **bug tracking** and sprint work using **ClickUp** and **Jira**, and maintained **version control** through **GitHub**, reducing incident resolution time by **35%**.
- Developed and executed **Unit Tests** and **Integration Tests** for backend services and API workflows, improving software quality, application reliability, and release stability.
- Deployed and supported highly available applications across **AWS** using **EC2**, **S3**, **RDS**, and **IAM**, maintaining **CI/CD Pipelines** through **GitHub Actions** and **AWS CodePipeline** to ensure reliable and scalable software releases.

ROWAN UNIVERSITY, NEW JERSEY | RESEARCH COLLABORATOR

JUNE 2024 - FEBRUARY 2025

- Designed and developed an automated exam evaluation workflow using **Python**, **FastAPI**, **MongoDB**, and **AWS**, reducing manual grading effort for professors and supporting assessment workflows for 1,000+ students per semester.
- Built scalable backend services and **REST APIs** using **FastAPI**, processing student submissions, grading workflows, and evaluation results for courses supporting 1,000+ students per semester.
- Containerized application components using **Docker**, enabling reproducible deployments, simplified environment management, and streamlined development and testing workflows.
- Designed and optimized **MongoDB** data models for storing exam submissions, grading results, and workflow metadata, supporting reliable processing of high-volume academic records.
- Collaborated with faculty members to gather requirements, improve grading workflows, and automate repetitive evaluation tasks, reducing turnaround time for exam result generation.
- Deployed and maintained research applications on **AWS**, leveraging **EC2**, **S3**, and cloud-native services to support scalable processing, secure data storage, and reliable access to grading services.

DOCUBYTE SOFTWARE SOLUTIONS, INDIA | SOFTWARE DEVELOPER

JUNE 2020 - AUGUST 2022

- Contributed to MedPlusMart, India's largest e-pharmacy serving 350,000+ daily customers, developing **Microservices** using **Java**, **Spring Boot**, and **ASP.NET Core** to support high-volume prescription management, order processing, and diagnostics booking workflows.
- Developed and enhanced **REST** and **SOAP APIs** in **ASP.NET Core** using the **.NET SDK** with **Entity Framework Core** and **SQL Server** for core pharmacy workflows including prescription upload and verification, click-and-pick order management, home delivery scheduling, and pharmacist authorization flows, with admin-facing dashboards built using **Angular** and **TypeScript**.
- Built scalable **HL7/FHIR APIs** using **Spring Boot**, enabling secure exchange of healthcare data and supporting interoperability across distributed healthcare systems.
- Implemented **HIPAA**-aligned system security practices across **Java** and **C#** services including audit logging, role-based access control using **ASP.NET Identity**, field-level encryption for prescription and patient data, and secure API authentication and authorization using **OAuth 2.0** and **JWT**-based token validation.
- Implemented automated testing strategies using **JUnit** and **NUnit** for backend services and healthcare workflows, improving software quality and reducing production defects.
- Designed and optimized **SQL Server** and **PostgreSQL** schemas supporting high-volume transactional data for **1,500+ pharmacy stores**, including inventory synchronization, order tracking, billing workflows, and **MedPlus Payback Points** rewards processing with real-time balance updates.
- Deployed and maintained healthcare microservices on **Azure App Services**, building **Azure DevOps CI/CD pipelines** with **Bitbucket** across **6 environments** (Alpha, Beta, Gamma, Production, Client Sandbox, Dev Sandbox) for automated builds and multi-environment releases, ensuring platform reliability for a high-traffic e-pharmacy serving **300+ cities** across India.

PROJECTS

- Watch Wave** - Built a full stack **React** and **JavaScript** streaming web application integrating **OpenAI API** for AI-powered recommendations, **Firebase** for authentication and real-time data, **Redux** for state management, and **TMDB API** for content discovery, delivering a seamless user experience.
- E-Wallet Express** - Designed and developed a **Microservices**-based e-wallet platform using **Java** and **Spring Boot**, leveraging **Kafka** message queues for asynchronous event-driven communication and **Redis** caching to improve scalability, performance, and fault tolerance.

EDUCATION

Rowan University	GPA: 3.63/4.0	SEPTEMBER 2022 - MAY 2024
<i>Master of Science in Computer Science</i>	Courses: Advanced Design & Algorithms, Advanced Database, Network Security, Functional programming	
Institute Of Aeronautical Engineering	GPA: 3.23/4.0	AUGUST 2018 - MAY 2022
<i>Bachelor in computer science and engineering</i>	Courses: Data Structures, E-Commerce, Web Technologies, Cloud Application Development	